

Clara Dehman

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Department of Physics, University of Alicante, 03690 Alicante, Spain

Education

Ph.D. in Physics

Nov 2019 — Nov 2023

Autonomous University of Barcelona, Spain

- *Thesis Title:* Unveiling the Physics of Neutron Stars: A 3D expedition into MAGneto-Thermal evolution in Isolated Neutron Stars with “MATINS”
- *Honors:* Cum Laude Distinction · International Mention · IEEC Best Thesis Award · SEA Thesis Award (Special Mention)

International Joint M.Sc. in Theoretical Nuclear Physics

Oct 2017 — Sep 2019

Erasmus Mundus NucPhys Program, European Union

Degrees Awarded:

- Laurea Magistrale in Fisica, *University of Padova, Italy* — 110/110 + Cum Laude
- Master de Sciences, Technologies, Santé, Mention Physique, *University of Caen Normandy, France*
- Máster Universitario Erasmus Mundus en Física Nuclear, *Spanish Consortium Universities* — Matrícula de honor

B.Sc. in Physics

Oct 2014 — Aug 2017

Lebanese University, Lebanon — Ranked 1st in Physics Department

Research Positions

Juan de la Cierva Postdoctoral Fellow

Mar 2024 — present

University of Alicante, Spain

- *Project:* Investigate the chiral magnetic effect, magnetic helicity, origin of magnetar magnetic fields, long-period transients, and sub-grid scaling.
- *Advisor:* Jose A. Pons

Transitional Postdoctoral Researcher

Nov 2023 — Feb 2024

Institute of Space Sciences (ICE-CSIC), Barcelona, Spain

- *Project:* Modeled 3D neutron star magnetic field evolution and benchmarked results against X-ray observations.
- *Advisor:* Nanda Rea

Visiting Ph.D. Fellow

Apr — Dec 2023

Nordita, Stockholm, Sweden

- *Project:* Studied the role of magnetic helicity in the inverse cascade of neutron star magnetic fields.
- *Advisor:* Axel Brandenburg

Doctoral Researcher

Nov 2019 — Nov 2023

Institute of Space Sciences, Barcelona, Spain

- *Project:* Developed MATINS, a 3D code for magneto-thermal evolution of isolated neutron stars; applied it to cooling ($\pm$ hyperons), magnetar outbursts, and interior–magnetosphere coupling using physics-informed neural networks.
- *Advisors:* Nanda Rea, Daniele Viganò, Jose A. Pons

Graduate Researcher

Feb — Sep 2019

University of Barcelona, Spain

Grade: 10/10 + Matrícula de Honor

- *Project:* Computed finite-temperature equations of state using the BCPM energy density functional for post neutron star mergers and late-stage proto-neutron stars.
- *Advisors:* Artur Polls, Xavier Viñas, Mario Centelles

Intern

Sep 2018 — Jan 2019

Laboratory of Corpuscular Physics, Caen, France

- *Project:* Developed metamodeling approaches for nuclear equations of state to interpret experimental data, focusing on the neutron star mass–radius relation and tidal deformability.
- *Advisors:* Francesca Gulminelli

Intern

Mar — Jun 2018

National Institute for Nuclear Physics, Legnaro Laboratories, Italy

- *Project:* Measured fusion cross sections of a ^{28}Si beam on a ^{100}Mo target at various energies to probe many-body quantum tunneling.
- *Advisors:* Giovanna Montagnoli, Alberto Stefanini

Fellowships & Research Fundings

COMPETITIVE FELLOWSHIPS & SCHOLARSHIPS AS PI

Total secured: €121,700, excluding €92,470 APOSTD Fellowship (declined)

- **Feb 2025 – Jan 2027:** *Juan de la Cierva Postdoctoral Fellowship*, Spanish Ministry of Science and Innovation — nationally competitive (10% success rate) — €74,734
- **Oct 2024 – Sep 2026:** *APOSTD Postdoctoral Fellowship*, Government of Valencia, Spain — €92,470 (declined in favor of the more prestigious Juan de la Cierva Fellowship).
- **Apr 2023 – Dec 2023:** *Nordita Visiting PhD Fellowship*, Nordita, Sweden — 80,000 kr
- **Oct 2017 – Sep 2019:** *EACEA Scholarship*, Erasmus Mundus Joint MSc Program, European Union — €37,000
- **Sep 2012 – Jun 2014:** *English Micro-Scholarship Program*, American Embassy, Lebanon — \$2,000

COLLABORATIVE RESEARCH GRANTS AS CONTRIBUTOR

Contributed to research grants totaling €2,944,000

- **Mar 2024 – Dec 2026:** *BLADES (Birth, Life, And Death of Extreme Sources)*, CIPROM 2022/13 Grant, Government of Valencia, Spain — PI: Jose A. Pons, €584,000
- **Mar 2024 – Dec 2026:** *Relativistic Computational Astrophysics*, ASFAE/2022/26 Grant, Government of Valencia, Spain — PI: Miguel A. Aloy, €300,000
- **Apr 2022 – Dec 2024:** *SGR-Cat 2021 (01269) Grant*, Government of Catalonia, Spain — PI: Vanessa Graber, €60,000

- **Nov 2019 – Feb 2024:** *Magnesia (The magnetar census and impact of highly magnetic neutron stars on the explosive and transient Universe)* ERC Consolidator Grant, European Research Council — PI: Nanda Rea, €2,000,000

Awards & Certificates

- **Dec 2024:** *IEEC Best Doctoral Thesis Award 2024* — €1,500 awarded to me, plus €1,000 allocated to my thesis supervisors as part of the prize.
- **May 2024:** *Special Mention, SEA Thesis Award 2024* — for outstanding research quality in a highly competitive category
- **Mar 2019:** *Certificate in Data Analysis and Machine Learning*, Michigan State University (USA) and University of Oslo (Norway)
- **Apr 2017:** *ERASMUS+ Joint Master Program (NucPhys)* — selected among top 20% of international applicants (CV and Interview)

Invited Talks & Seminars (17 Total)

- **Jun 2026:** *Review Talk*, Frontier Research in Astrophysics - V, Palermo, Italy
- **Apr 2026:** *Strong Gravity Seminar*, Perimeter Institute, Waterloo, Canada
- **Apr 2026:** *Seminar*, Canadian Institute for Theoretical Astrophysics, Canada
- **Jun 2025:** *Talk*, Fòrum IEEC 2025, Casa Convalescència (UAB), Barcelona, Spain
- **May 2025:** *Talk*, Extreme Physics of Neutron Star Interiors, Princeton University, USA
- **May 2025:** *Review Talk*, Workshop on Modern Equation of State and Spectroscopy in Neutron Stars, Alcalá University, Spain
- **Feb 2025:** *Talk*, Breaking New Ground in Supernova Physics, Fukuoka University, Japan
- **Dec 2024:** *Review Talk*, IReNA-INT Joint Workshop on Thermal and Magnetic Evolution, University of Washington, USA
- **Sep 2024:** *Talk*, Frontier Workshop in Astrophysics IV, Palermo, Italy
- **Jun 2023:** *Astrophysics Seminar*, University of Valencia, Spain
- **May 2023:** *Astrophysics Seminar*, Nordita, Sweden [[Link](#)]
- **Jun 2022:** *Astrocoffee Seminar*, Goethe University, Germany [[Link](#)]
- **May 2022:** *Seminar*, NucPhys Master Program, GANIL, France
- **Dec 2021:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain [[Link](#)]
- **Nov 2021:** *Astrophysics Seminar*, University of Alicante, Spain
- **Dec 2020:** *Talk*, Stars & Compact Objects Meeting, Flatiron Institute, New York, USA
- **Mar 2019:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain

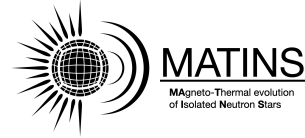
Contributed Talks & Seminars (11 Total)

- **Jun 2025:** *Talk*, EAS Annual Meeting, University College Cork, Ireland
- **Jul 2024:** *Talk*, 10th International Conference on Quarks and Nuclear Physics, University of Barcelona, Spain
- **Jun 2024:** *Talk*, XMM-Newton Workshop 2024, European Space Agency, Madrid, Spain
- **May 2024:** *Special Seminar*, Institute of Space Sciences, Barcelona, Spain
- **Apr 2024:** *Astrophysics Seminar*, University of Alicante, Spain
- **May 2022:** *Talk*, Pharos Conference, University of La Sapienza, Rome, Italy
- **Mar 2022:** *Seminar*, Institute of Space Sciences, Barcelona, Spain [[Link](#)]

- **Nov 2021:** *Talk*, IAU Symposium 363 (online)
- **Jun 2021:** *Talk*, EAS Annual Meeting (online)
- **Nov 2020:** *Seminar*, Institute of Space Sciences, Barcelona, Spain [Link]
- **Mar 2020:** *Talk*, PHAROS Conference, Patras, Greece (canceled due to COVID-19)

Software Development & Open Access Tools

Since Sep 2020: Lead developer of MATINS — a 3D code for the MAgneto-thermal evolution in Isolated Neutron Stars — [MATINS webpage](#)



- *Open access:* github.com/ice-csic-astroexotic/MATINS
- *Lead developers:* **C. Dehman**, D. Viganò, S. Ascenzi
- *Implementation:* Written in Fortran 90 with OpenMP parallelization, optimized for Intel compilers (better performance than GNU).

Supervision

PhD Students — *Co-supervisor; University of Alicante, Spain*

- Joan Llorens Ripoll — Ambipolar diffusion in neutron star cores (since Mar 2025)

MSc Students — *Co-mentor; Institute of Space Sciences, Spain*

- Riccardo Ricci — Magnetar activity and crustal failures (since Oct 2025)
- Marco Ermolli — Impact of microphysics on neutron star cooling (Jan – Jul 2021)

BSc Students — *Main supervisor; University of Alicante, Spain*

- Ahlam Makboul — Glitch timing and magnitude of PSR J0537–6910: probing crust–magnetic field interactions (since Oct 2025)

Teaching (112 Hours Total)

Instructor — **Introduction to Modeling in Physics**

University of Alicante, Spain

Editions 2024 & 2025

45 hr — 95 BSc students

Instructor — **Nuclear and Particle Physics**

University of Alicante, Spain

Editions 2024 & 2025

54 hr — 112 BSc students

Lecturer — **Stellar Astrophysics**

University of Alicante, Spain

Jan — Mar 2025

12 hr — 14 BSc students

Invited Lecturer — **Research in Computational Astrophysics**

University of Alicante, Spain

May 2025

1 hr — 10 MSc students

Scientific Community Service, Outreach & Visibility

SCIENTIFIC MEETING ORGANIZATION

SOC: Scientific Organizing Committee; LOC: Local Organizing Committee

- **Sept 2026:** *SOC & LOC*, 22nd Pencil Code User Meeting, University of Alicante, Spain
- **Nov 2022:** *LOC*, Exploring the Hot and Energetic Universe: the 3rd scientific conference dedicated to the Athena X-ray observatory, Barcelona, Spain [Link]
- **May 2022:** *LOC*, PHAROS Final Conference 2022: The multi-messenger physics and astrophysics of neutron stars, University of La Sapienza, Rome, Italy [Link]

REVIEWING & EVALUATION ACTIVITIES

- **Jan 15, 2026:** *PhD committee member* – Juan A. Gil Granados, Univ. of Barcelona, Spain
- **Since Jun 2025:** *Expert evaluator for PhD proposals* (2 proposals evaluated), Valencian Agency for Assessment and Prospective, Spain
- **Since 2022:** *Reviewer* for Nature Astronomy; Journal of Physics G; Monthly Notices of the Royal Astronomical Society; Frontiers in Astronomy and Space Sciences; Classical and Quantum Gravity

OUTREACH ACTIVITIES

- **Feb 2026:** *Invited Outreach talk* — International Day of Girls and Women in Science (University of Alicante, Spain)
- **Nov 2022:** *Astronomical Observation* — Contributed to organizing an astronomical observation (Autonomous University of Barcelona, Spain; [\[Link\]](#))
- **Sep 2022:** *Research Night* — Observation night for school children (Igualada, Spain)
- **Feb 2022:** *Outreach talk* — International Day of Girls and Women in Science (ICE-CSIC, Spain; [\[Link\]](#))
- **2021–2023:** *Magnet ICE Outreach* — Collaborated with teachers at Gabriel Castellà i Raich in creating innovative science projects to promote student integration in a school facing segregation issues (Igualada, Spain).

PRESS RELEASES (14 TOTAL)

- **Jul 2024:** *Descubren tres estrellas de neutrones jóvenes inusualmente frías para su edad*, [europapress](#)
- **Jul 2024:** *Investigadores de la UA detectan estrellas de neutrones inusualmente frías*, [TODOAlicante](#)
- **Jul 2024:** *Investigadores de la UA participan en el descubrimiento de tres estrellas de neutrones demasiado frías para su edad*, [Actualidad Universitaria](#)
- **Jun 2024:** *NASA's Chandra Peers Into Densest and Weirdest Stars*, [NASA](#)
- **Jun 2024:** *Too young to be so cool: lessons from three neutron stars*, [ESA](#)
- **Jun 2024:** *Un equip d'astrònoms de l'IEEC descobreix tres estrelles de neutrons massa fredes per a la seva edat*, [IEEC](#)
- **Jun 2024:** *Troppo giovani per essere così fredde*, [MEDIA INAF](#)
- **Jun 2024:** *A team of ICE-CSIC astronomers discovers three neutron stars too cold for their age*, [ICE-CSIC](#)
- **Dec 2022:** *Astronomers observe outburst of the young magnetar Swift J1818.0–1607*, [Phys Org](#)
- **Jun 2020:** *NASA discovers youngest 'cosmic baby' neutron star*, [Independent](#)
- **Jun 2020:** *Astronomers just discovered the youngest ever 'baby' dead star*, [CNET](#)
- **Jun 2020:** *Astronomers find baby pulsar – a rare magnetar – born just 240 years ago*, [Astronomy Now](#)
- **Jun 2020:** *Astronomers discover youngest magnetar ever*, [Sci News](#)
- **Jun 2020:** *Hallan estrella 'bebé' que explica origen de las explosiones en el universo*, [La Vanguardia](#)

INVITED INTERVIEW

- **Jun 2024:** *Invited Interview* at University of Elche, Spain [[Link](#)]
- **Dec 2023:** *Invited Interview* at ICE-CSIC, Barcelona, Spain [[Interview](#); [Newsletter](#)]

Languages, Memberships & Trainings

COLLABORATIONS & MEMBERSHIPS

SCALES European COST Action CA24139 ([since Nov 2025](#)) • NewAthena Science Community ([since Mar 2025](#)) • Einstein Telescope Collaboration ([since Sep 2022](#)) • AIHUB, Artificial Intelligence Initiative of Spain's National Research Council ([since Mar 2022](#)) • Pharos European COST Action CA16214 (2019 – 2022)

LEADERSHIP TRAINING

Project Citizen (AMIDEAST & American Embassy, Lebanon, 2014) • Intel Youth Innovation Camp (INJAZ, 2013) • Personal Strategic Planning (IAPTL, 2013) • Leadership Program (AMIDEAST & INJAZ, 2012)

LANGUAGES

[Arabic](#) (C2, mother tongue) • [English](#) (C2, fluent; Advanced Cambridge 183/210, TOEFL 805/900) • [French](#) (C2, fluent) • [Spanish](#) (B2, intermediate) • [Italian](#) (B2, intermediate)